

TECHNICAL OVERVIEW

CAD Repair System

Evidence-bounded geometry repair before CFD and FEA meshing

CRS locates defects in CAD topology, attempts narrow kernel-level repairs, and keeps a change only when independent re-inspection shows measurable progress inside conservative geometry limits.

Prepared by Hexera | October 2026 | Client briefing

SYSTEM OVERVIEW

Repair is a measured transaction, not a cleanup filter

Simulation workflows often fail before solver setup because imported CAD is topologically invalid, internally inconsistent, or burdened by geometry much smaller than the intended mesh scale. CRS sits between CAD intake and mesh generation. It diagnoses the B-rep, applies only an eligible repair, then judges the written result rather than trusting the repair operation's own success flag.

01	02	03	04	05
Inspect	**Plan**	**Execute**	**Measure**	**Promote**
Read B-rep and localize defects	Choose the narrowest eligible repair	Run the CAD kernel on a snapshot	Re-scan geometry and compare caps	Stage repaired CAD for the mesh engine

Core guarantee

The customer upload remains immutable. If repair is unsupported, unsafe, inconclusive, over budget, or rejected by engine staging, the repaired candidate is not promoted.

What the system is designed to improve

- B-rep validity and consistency before surface or volume meshing.
- Meshability of assemblies that contain local topology defects, sliver geometry, or broken boundaries.
- Repeatability: defect identity, attempted operation, measurements, and verdict are recorded for each run.
- Time to a usable mesh by avoiding manual CAD cleanup when the repair is mechanically defensible.

What it does not claim

CRS does not infer engineering intent, certify boundary conditions, validate material models, or prove a CFD/FEA solution is physically correct. It prepares and verifies geometry for the next stage. Solver validation remains an engineering responsibility.

CURRENT CAPABILITIES

Defects CRS detects and can address

The table distinguishes a repair operation from a guaranteed outcome. A defect is repaired only when it is localized, an operation claims it, the result improves on re-scan, and all safety checks pass.

Defect class	CRS response today	Boundary
Invalid B-rep and orientation failures	Localized whole-shape repair; result is re-inspected.	Kept only when validity improves without breaching shape caps.
Wire gaps and face-boundary inconsistency	Repairs wire ordering, p-curves and face boundaries.	Target must be identified; no blind topology rewrite.
Tiny or degenerate edges	Attempts local edge and boundary healing.	Warnings may remain when removal would alter intended detail.
Sliver faces	Deployment-controlled sweep of measured sub-tolerance faces.	Every removed face is counted and bounded; otherwise rollback.
Self-intersecting wires/faces	Targeted healing on kernel-identified geometry.	Deployment-controlled; must reduce error severity on re-scan.
Open shells and missing faces	Sews coincident boundaries; may patch an isolated bounded opening.	Hole filling is guarded and refuses ambiguous through-passages.
Non-manifold, intent-sensitive or staging defects	Detects and routes with evidence.	Not automatically rewritten when engineering intent is uncertain.

Conservative by design

CRS may leave non-blocking warnings in place. Removing every small feature would be defeaturing, which can change the physics of a model. The system prioritizes simulation readiness without pretending that all fine geometry is disposable.

Typical dispositions

REPAIR	Localized defect with a bounded operation and measurable progress.
MESH AS IS	No blocking defect, or only accepted warnings for the chosen workflow.
REVIEW	The geometry is understood but the right change depends on engineering intent.
REFUSE	The candidate damages topology, exceeds a cap, or fails independent staging.

SAFETY ARCHITECTURE

Every candidate must earn promotion

A kernel reporting success is not sufficient. CRS evaluates each candidate against both local progress and cumulative geometry limits measured from the original upload.

01	**Target integrity**	The defect must still exist at the identified entity. Stale or untraceable targets are refused.
02	**Topology sanity**	A candidate may not lose solids, erase undeclared faces, leave no meshable faces, or turn a valid body invalid.
03	**Geometry bounds**	Tolerance growth, overall size, surface area, and volume are compared with the original and the operation's declared change.
04	**Independent progress**	The candidate is re-inspected. The target defect must disappear or materially improve without introducing a new warning-or-worse defect.
05	**Engine staging**	The repaired file must stage for the selected mesh engine before it can replace the original in the run.

Deterministic operating envelope

- Repair rules execute in a fixed priority order from narrow local operations to broader fallback operations.
- Attempt count, elapsed time, and consecutive non-progress outcomes are bounded.
- Every failed or rolled-back operation is retained as evidence and is not silently retried.
- Optional model guidance can propose a repair, but the CAD kernel and the same deterministic checks still decide whether it is kept.

Rollback behavior Operations work from snapshots. A declined, failed, or rejected attempt restores the prior accepted shape. No rejected candidate becomes customer output.

FORMULA 1000 EVIDENCE

Full vehicle body: blocking errors removed, solids preserved

Hexera evaluated an open Formula 1000 race-car body distributed by Ludemann Engineering. The source file was used as received for the before case. CRS produced a separate repaired STEP file for the after case.

171	8 -> 8	INVALID -> VALID	16.87 -> 7.99 s
findings before	solids preserved	kernel verdict	cfMesh staging

Measure	Before CRS	After CRS	Interpretation
OpenCASCADE validity	Invalid	Valid	Error-level topology was resolved.
Solids	8	8	No solid body was lost.
Faces	421	419	Bounded cleanup changed topology without changing extents.
Reported findings	171 total	162 warnings	CRS resolved the blocking errors; it did not claim every warning disappeared.
cfMesh CAD staging	16.87 s	7.99 s	Both staged; repaired input completed in about half the time in this run.
Surface triangles	147,545	153,117	The mesher consumed the repaired topology and produced a different surface discretization.

Defects observed before repair

The initial inspection reported two invalid B-rep findings, two self-intersecting-wire findings, 132 sliver-face findings, and 35 tiny-edge findings. CRS resolved the blocking invalidity and removed four measured sliver faces during the accepted sequence. It left 162 warning-level findings rather than over-defeaturing the body.

How to read this result

This is evidence of CAD repair and mesh-engine readiness on one public vehicle model. It is not a benchmark across all CAD, and the surface staging result is not a completed CFD or FEA solve.

Model attribution: Formula 1000 Race Car Design Files by Ludemann Engineering. Source and license: github.com/LudemannEngineering/Formula-1000-Race-Car. The repaired file is a Hexera-generated modification used for evaluation. Ludemann Engineering does not endorse Hexera or this result. Independent suitability verification is required by the source license and by normal simulation practice.

DEPLOYMENT FIT

Where CRS fits in a simulation workflow

CRS is intended as a pre-meshing quality gate for CAD-driven CFD and FEA. It is most valuable when teams receive geometry from multiple CAD systems, cannot standardize authoring practices, or spend repeated engineering time converting almost-usable files into meshable bodies.

Companion part result

On the Formula front suspension rocker, independent inspection measured 12 warnings before and 8 after repair. CRS reduced sliver faces from four to one, preserved the single solid, and kept the B-rep valid. Remaining warnings were reported rather than hidden.

Best-fit use cases

- External-aerodynamics and conjugate-heat-transfer preparation.
- Mechanical assemblies with import or translation damage.
- Batch simulation pipelines where manual cleanup blocks throughput.
- Engineering services that need an auditable before/after record.

Evidence delivered with a run

- Original and promoted geometry identity.
- Located defects with severity and geometric context.
- Each attempted operation and keep/rollback verdict.
- Post-repair measurements and remaining findings.
- Selected mesh-engine staging outcome.

Acceptance criteria for a production evaluation

- Use representative customer CAD rather than synthetic-only fixtures.
- Define the target mesh engine, scale, and simulation objective before judging warnings.
- Compare original and repaired CAD for body count, extents, tolerances, area/volume, and expected passages.
- Run the normal mesh quality checks and at least one solver smoke case on promoted geometry.
- Escalate intent-sensitive changes to an engineer; do not tune caps until an unsafe repair passes.

Bottom line

CRS automates geometry repair where the defect, operation, and proof of improvement are all explicit. It refuses to convert uncertainty into a clean-looking but untrustworthy model.

Hexera

CAD intelligence and deterministic mesh preparation for engineering simulation.